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FAOSTAT ANALYTICAL BRIEF 122

Temperature change statistics 1961–2025

Global, regional and country trends

>> FAO Statistics Division

HIGHLIGHTS

→ **The 2025 global mean annual temperature change on land was 1.9 °C compared with the 1951–1980 baseline, making it the second warmest year on record after 2024 (2.1 °C). The years 2023, 2024 and 2025 were the three warmest consecutive years ever recorded.**

→ **January 2025 was the warmest month ever recorded. Each month of 2025 exceeded 1.3 °C above the 1951–1980 climatology.**

→ **In 2025, Europe (2.7 °C), the Americas (1.9 °C) and Asia (1.8 °C) exceeded the Paris Agreement threshold of 1.5 °C, while Oceania (1.4 °C) and Africa (1.3 °C) remained below it.**

→ **At the subregional level, warming was strongest in Central Asia (3.1 °C), Eastern Europe (2.8 °C), and Northern America and Northern Europe (2.5 °C). The lowest values were observed in Melanesia, Polynesia and Southern Africa (0.9 °C), followed by Middle Africa (1.1 °C).**

→ **Nearly half of all countries and territories (109 out of 223) experienced mean annual warming greater than 1.5 °C in 2025, and 55 exceeded 2.0 °C. The Svalbard and Jan Mayen Islands (Norway) recorded the highest anomaly worldwide for the fifth year in a row, at 3.9 °C.**

→ **In 2025, areas exposed to warming greater than 1.5 °C represented around half of the world population (4.1 billion people) and half of the agricultural land area (2.6 billion hectares).**

TEMPERATURE CHANGE

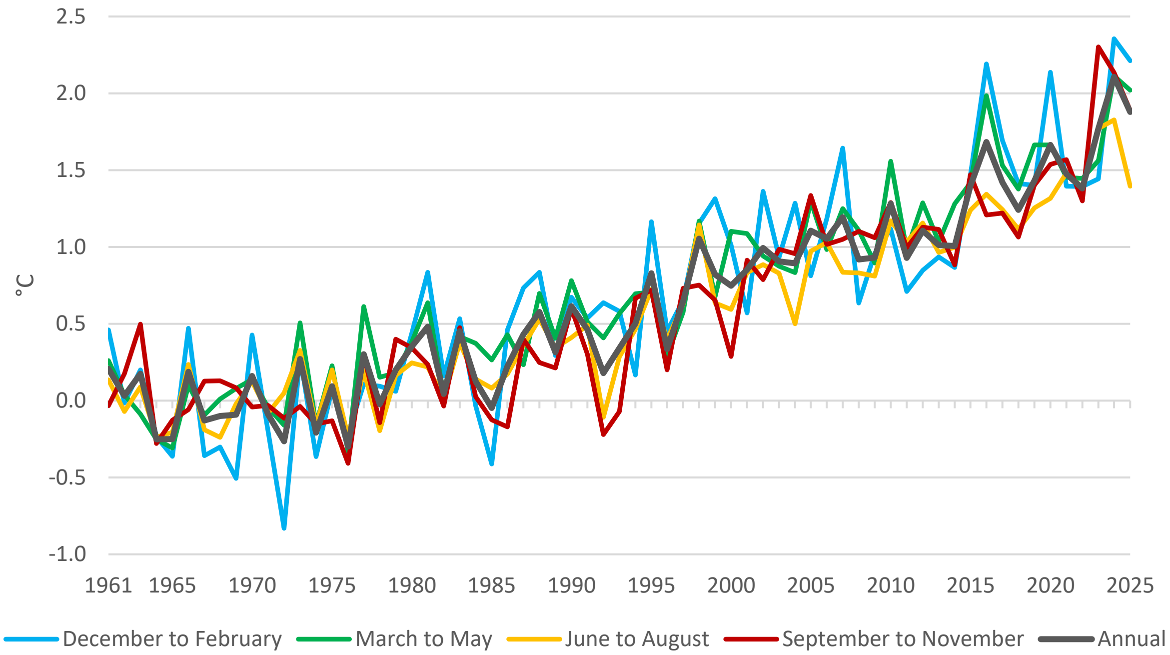
INTRODUCTION

Anthropogenic greenhouse gas emissions cause global warming. Their abatement is at the core of current international policy commitments to reduce future projected and increasingly observed climate impacts on livelihoods throughout the world, and in particular to safeguard food production (IPCC, 2023). The Paris Agreement seeks to limit warming to below 2 °C and possibly to no more than 1.5 °C. The FAOSTAT Temperature change on land dataset allows to monitor the observed warming trends on land at the country, regional and global level, with annual, seasonal and monthly means, to help gauge expected risks to food and agriculture. The data are disseminated for the period 1961–2025 for 197 countries and 31 territories. They are produced in collaboration with the NASA Goddard Institute for Space Studies (NASA–GISS). Temperature change is computed with respect to a reference climatology corresponding to the period 1951–1980 (see Explanatory Notes).

GLOBAL

In 2025, the global mean annual land temperature anomaly was 1.9 °C high above the climate normal of the period 1951–1980, making it the second-warmest year on record after 2024 (+2.1 °C). This makes 2023, 2024 and 2025 the three warmest consecutive years on record since 1880, the beginning of direct measurements, and reinforces the increasing warming of the past decade. Figure 1 illustrates trends in the world mean annual and seasonal temperature change on land. It shows the near linear warming trends observed since the mid-1980s, with the last decade having reached a "new normal" around +1.5 °C, representing the Paris Agreement threshold that humanity should not cross to avoid catastrophic climate change. The figure also shows how seasonal temperature change has even already surpassed +2.0 °C in different seasons over the last decade.

Figure 1: Mean annual and seasonal temperature change on land

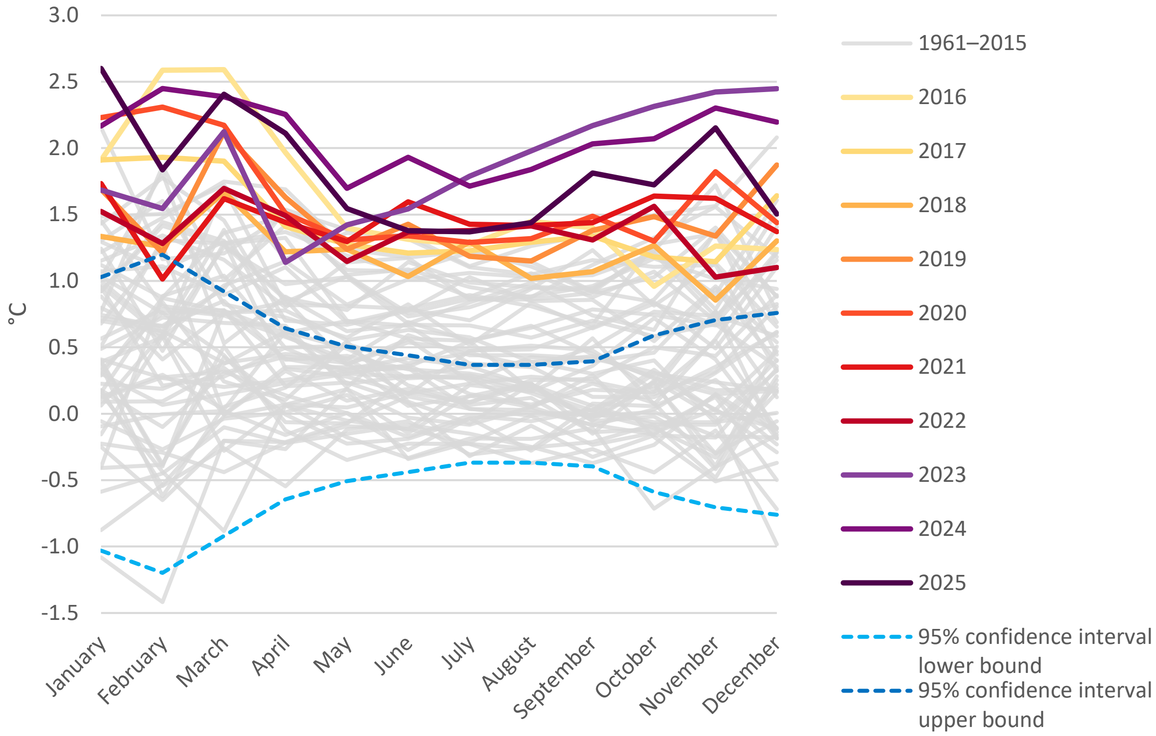


Note: The concepts of winter, spring, summer and autumn represent seasons that occur during opposite months in the two hemispheres. For instance, December, January and February correspond to winter in the northern hemisphere and summer in the southern hemisphere.

Source: FAO. 2026. FAOSTAT: Temperature change on land. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/ET. Licence: CC-BY-4.0.

Figure 2 illustrates the extent to which 2025 ranks among the warmest years on record: January was the warmest ever, and March to November were among the five warmest on record, with an average temperature change on land of at least 1.3 °C. In fact, each month of each year in the last decade was much warmer than the corresponding 1951–1980 baseline, with the levels of observed temperature change well above the 95 percent confidence interval upper limit of that reference climatology for all months but one. The figure represents the concept of the "loaded dice", first put forward by NASA climatologist Jim Hansen in the 1990s (Hansen, 2012). The visible effects of climate change can be seen as cumulatively increasing the probability that future years – and months in the year – break records compared to unperturbed climate.

Figure 2: Mean monthly world temperature change on land by year

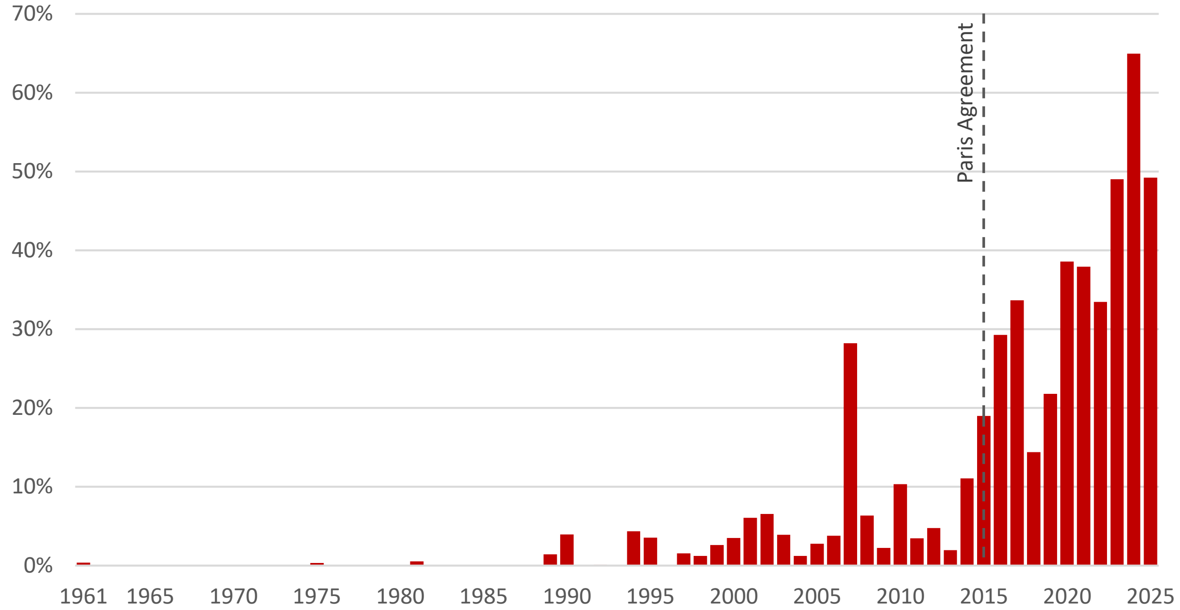


Source: FAO. 2026. FAOSTAT: Temperature change on land. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/ET. Licence: CC-BY-4.0.

Temperature change data in each country and territory were analysed as normal, warm or cold anomalies based on their distance from the 1951–1980 mean (i.e. within or outside the 95 percent confidence interval). In 2025, mean annual temperatures were warmer than normal in 225 countries and territories, and much warmer than normal (99 percent confidence interval) in 220. By comparison, in 2000, mean annual temperatures were warmer than normal in 168 countries and territories, and much warmer than normal in 85.

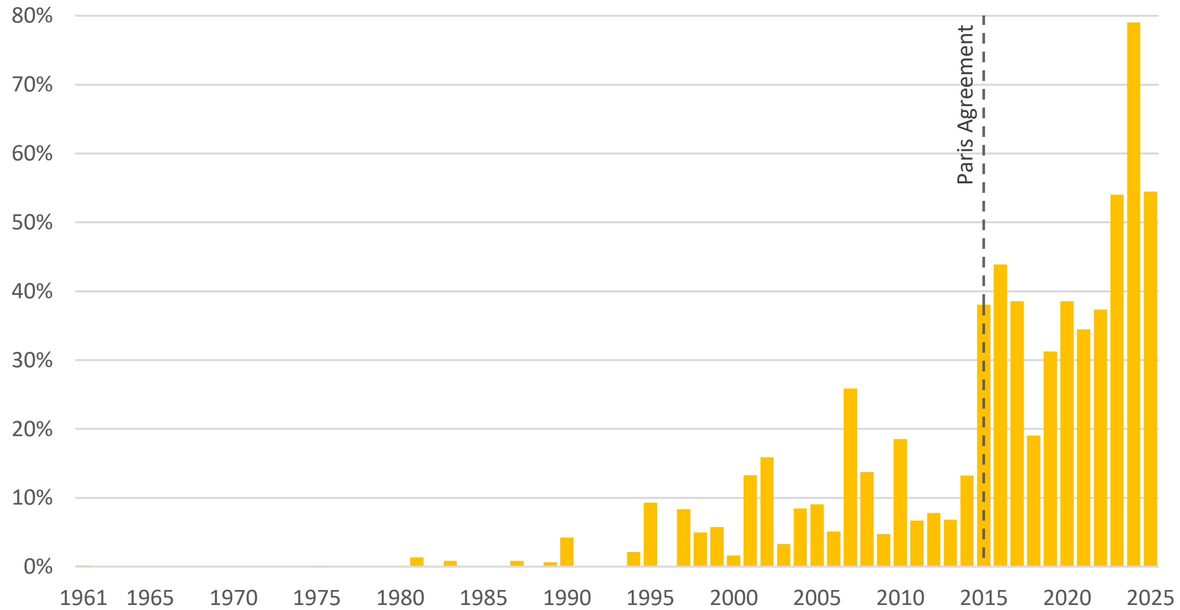
In 2025, the observed annual warming was greater than 1.5 °C in 109 out of 223 countries and territories, representing nearly half the world population (4.1 out of 8.2 billion people) (Figure 3). Furthermore, countries with warming above 1.5 °C represented 54 percent of the agricultural land area (2.6 out of 4.8 billion hectares) (Figure 4). This represents a major shift compared with the 2000s, since these shares were below 4 percent and 2 percent, respectively.

Figure 3: Share of the world population living in countries with warming above 1.5 °C



Source: Based on FAO. 2026. FAOSTAT: Temperature change on land. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/ET. Licence: CC-BY-4.0 and FAO. 2026. FAOSTAT: Annual population. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/OA. Licence: CC-BY-4.0.

Figure 4: Share of the total agricultural land area of countries with warming above 1.5 °C

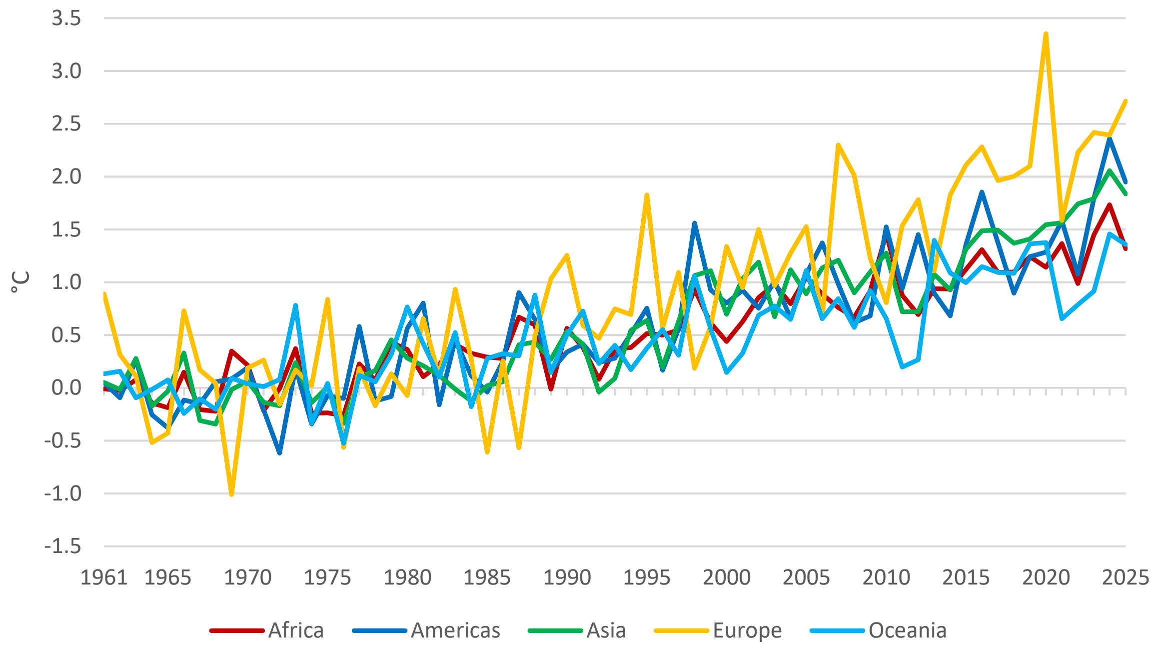


Source: Based on FAO. 2026. FAOSTAT: Temperature change on land. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/ET. Licence: CC-BY-4.0 and FAO. 2025. FAOSTAT: Land Use. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/RL. Licence: CC-BY-4.0.

REGIONAL

Europe recorded the largest temperature increase (2.7 °C) among regions, followed by the Americas (1.9 °C) and Asia (1.8 °C), with 2025 being the ninth of the last ten years with warming above 2.0 °C in Europe (Figure 5). Central Asia (3.1 °C), Eastern Europe (2.8 °C), Northern America and Northern Europe (both with 2.5 °C) were the subregions with the largest temperature change. Europe (and Eastern Europe) was the only region that experienced recent temperature change over 3.0 °C (in 2020). Oceania (1.4 °C) and Africa (1.3 °C) had substantial warming in 2025, below the 1.5 °C Paris threshold.

Figure 5: Mean annual temperature change on land by region



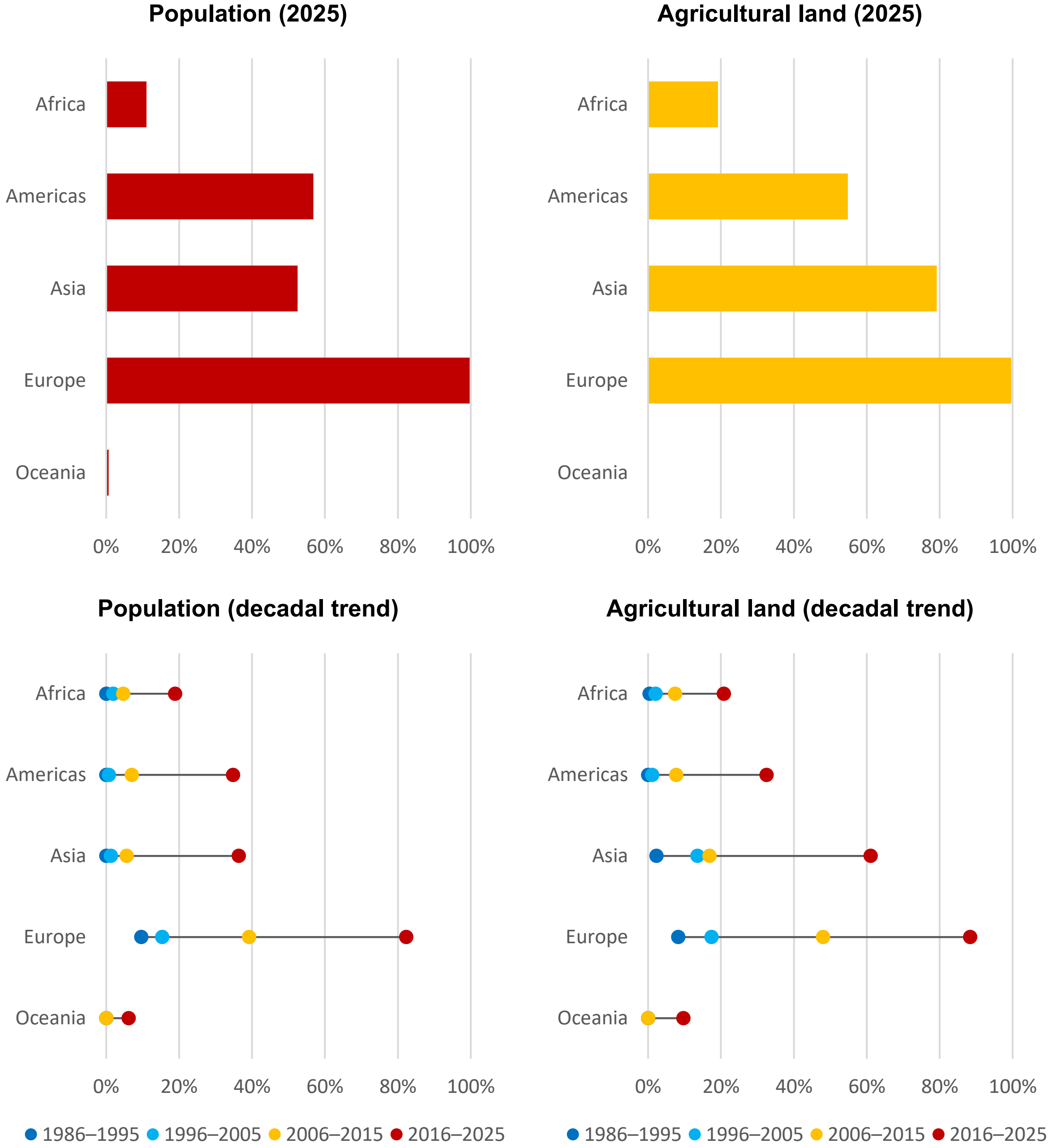
Source: FAO. 2026. FAOSTAT: Temperature change on land. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/ET. Licence: CC-BY-4.0.

In 2025, the population living in countries with warming above 1.5 °C was 2.5 billion in Asia, 740 million in Europe, 600 million in the Americas, 170 million in Africa and 300 000 in Oceania. In terms of risk, Europe was the most exposed, with almost 100 percent of its population living in countries with temperature change above 1.5 °C, followed by the Americas (57 percent), Asia (53 percent), Africa (11 percent) and Oceania (less than 1 percent) (Figure 6).

Agricultural land area in countries experiencing warming above 1.5 °C was the largest in Asia (1 300 million hectares), followed by the Americas (620 million hectares), Europe (450 million hectares), Africa (230 million hectares), and Oceania (less than 0.2 million hectares). In relative terms, Europe was the most exposed region, with nearly 100 percent of its agricultural land area in countries with warming above 1.5 °C, followed by Asia (79 percent), the Americas (55 percent), Africa (19 percent) and Oceania (less than 1 percent) (Figure 6).

The high values observed in 2025 should be put into perspective considering the trend observed over the last four decades. The share of population and agricultural land in countries where temperature change over land exceeded 1.5 °C has been doubling every decade in almost all regions.

Figure 6: Share of countries with warming above 1.5 °C by region

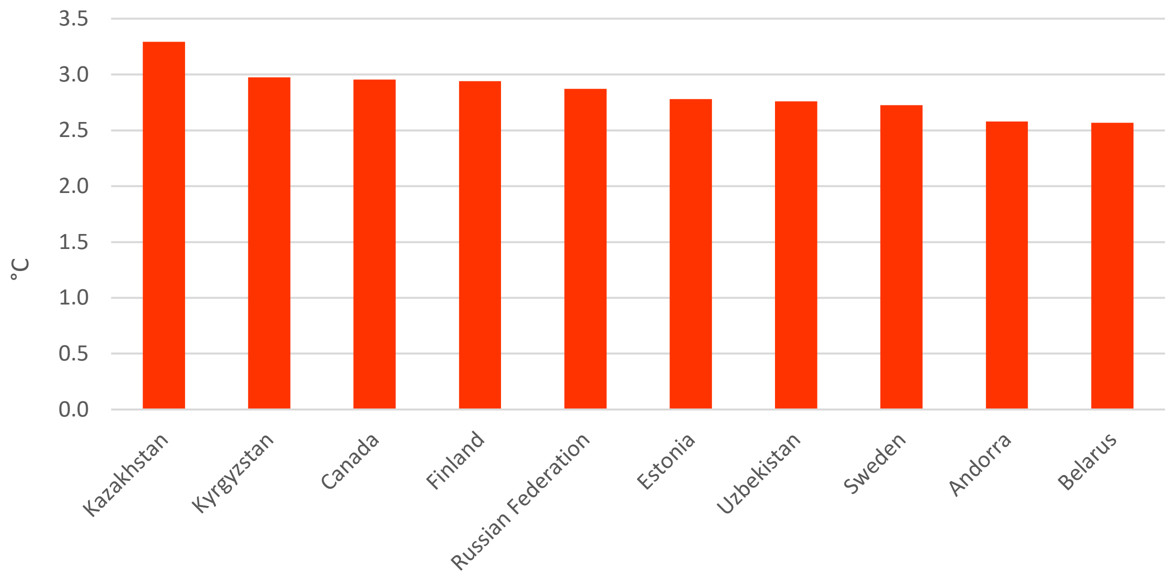


Source: Based on FAO. 2026. FAOSTAT: Temperature change on land. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/ET. Licence: CC-BY-4.0, FAO. 2026. FAOSTAT: Annual population. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/OA. Licence: CC-BY-4.0 and FAO. 2025. FAOSTAT: Land Use. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/RL. Licence: CC-BY-4.0.

COUNTRY

In 2025, the Svalbard and Jan Mayen Islands (Norway) recorded the highest warming globally for the fifth time in a row, reaching 3.9 °C above the 1951–1980 baseline. Most of the countries with the strongest warming were located across Northern and Eastern Europe, Central Asia, and Northern America, with values ranging from 3.3 °C in Kazakhstan to around 2.6–3.0 °C in Kyrgyzstan, Canada, Finland, the Russian Federation, Estonia, Uzbekistan, Sweden, Andorra and Belarus. Nearly half of all countries and territories experienced warming above 1.5 °C, and one in four exceeded 2.0 °C, confirming that the upper tail of the warming distribution expanded further in 2025 across high-latitude and continental regions.

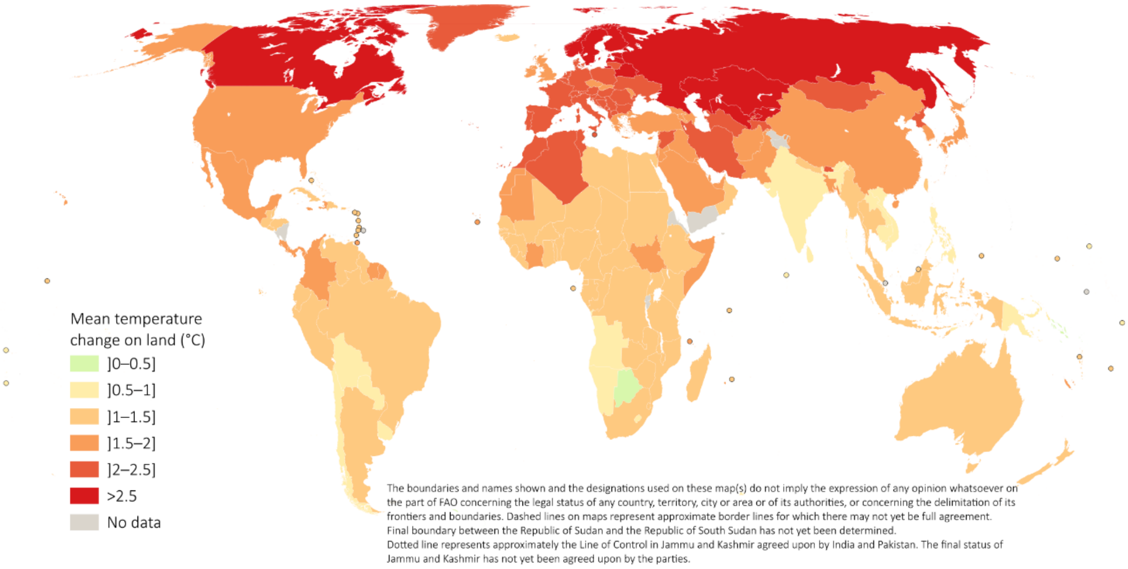
Figure 7: Countries with largest mean annual temperature change over land (2025)



Source: FAO. 2026. FAOSTAT: Temperature change on land. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/ET. Licence: CC-BY-4.0.

In contrast with the strong warming observed across most of the world in 2025, a small group of countries and territories experienced comparatively low temperature anomalies, generally below 1.0 °C. These are mainly tropical islands, oceanic territories and parts of the southern hemisphere, where large marine influences moderate annual temperature variability. The lowest values were recorded in the Solomon Islands (0.3 °C), South Georgia and the South Sandwich Islands (0.4 °C), Botswana (0.4 °C) and the Marshall Islands (0.5 °C), followed by Uruguay, the Plurinational State of Bolivia, Paraguay, Chile (0.6–0.8 °C). Many islands in the Pacific and Indian Oceans (e.g. Cook Islands, Tonga, French Polynesia, Maldives, Samoa) had warming ranging 0.8–1.0 °C, which is consistent with strong ocean buffering effects. The Lao People's Democratic Republic and Viet Nam (both with 0.8 °C) similarly experienced lower anomalies for 2025, reflecting the moderating role of humid tropical climates, dense vegetation and regional monsoon influences. Overall, the countries least affected in 2025 are characterized by maritime climates, equatorial or subtropical latitudes, and reduced continental amplification, leading to the lowest warming levels globally (Figure 8).

Figure 8: Observed mean annual temperature change (2025)



Source: FAO. 2026. FAOSTAT: Temperature change on land. [Accessed March 2026]. https://www.fao.org/faostat/en/#data/ET. Licence: CC-BY-4.0.

EXPLANATORY NOTES

The FAOSTAT Temperature change on land domain disseminates statistics of temperature change measured over land by country, with annual updates. The current database covers the period 1961–2025. Statistics are available for 237 countries and territories (223 countries and territories for 2025). They are available for monthly, annual and seasonal (winter, spring, summer and autumn) temperature anomalies, i.e. temperature change with respect to a reference climatology corresponding to the period 1951–1980. Data are also available for regional aggregates and special groups, including those relevant to the United Nations Framework Convention on Climate Change.

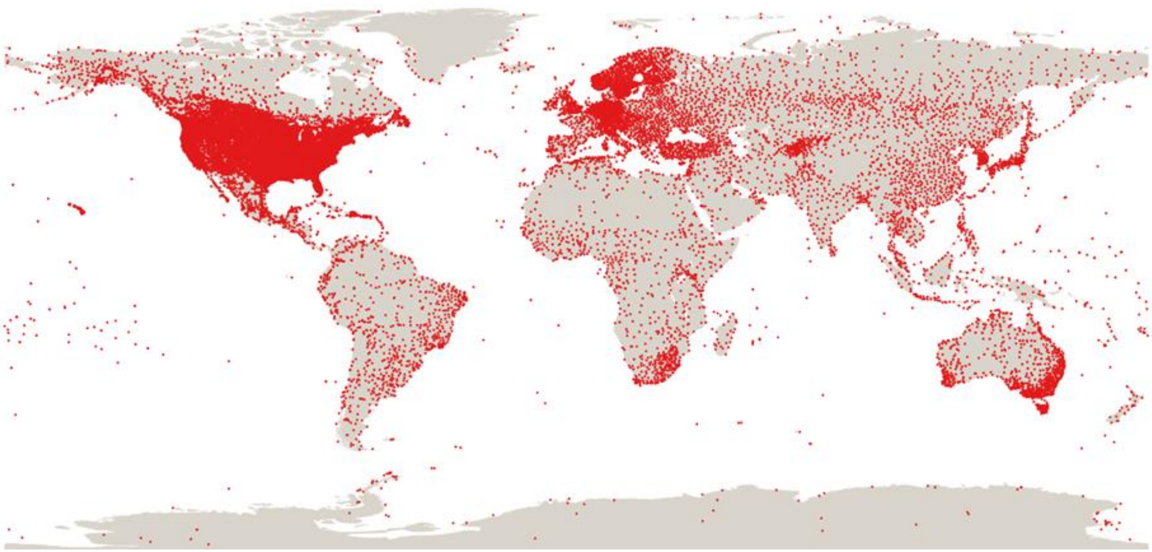
For each country or territory, and for each temperature variable considered, the mean and standard deviation σ of the 1951–1980 climatology was computed and disseminated in the dataset – except in cases with less than 20 available records in the time series. The standard deviation (σ) of the climatological mean annual, seasonal and monthly temperatures represents the natural interannual variability of that variable, by country. Warmer (or colder) temperature changes were defined as deviations exceeding 2 σ, hence outside the 95 percent confidence interval (CI). Similarly, much warmer (or much colder) temperature anomalies were those exceeding 3 σ, or outside the 99.7 percent CI. Conversely, temperature deviations within 2 σ from the climatological mean were taken to represent the interannual variability of the reference climatology, specifically ± 0.4 °C for the world; ± 0.4 °C for Africa; ± 0.5 °C for Asia; ± 0.6 °C for the Americas and Oceania; and ± 0.9 °C for Europe (95 percent CI).

The FAOSTAT data are based on the publicly available GISTEMP data, the Global Surface Temperature Change data distributed by the National Aeronautics and Space Administration Goddard Institute for Space Studies (NASA–GISS), with information from 1880 onwards. The original GISTEMP analysis generates a set of gridded values (resolution 500 km) from the Global Historical Climatology Network (GHCN v4), composed of continuously updated temperature data from over 26 000 meteorological stations on land around the globe (Figure 9). A finer grid (resolution 250 km) was prepared by NASA-GISS for the purpose of the FAOSTAT dataset, excluding ocean data, and subsequently aggregated at the country level using the FAO Global Administrative Unit Layers (GAUL). The FAOSTAT methodology includes reconstructing the time series to account for the administrative changes that occurred since 1961 (e.g. the split of the Soviet Union in 1991 or the separation of the former Sudan in 2011). Finally, country statistics were area-weighted to compute regional aggregates, using country area data from the FAOSTAT Land Use dataset as weights.

According to NASA, compared to statistics of absolute temperature value, a dataset based on temperature anomalies is more stable and coherent in the face of variations in coverage in space and time of the meteorological stations that underlie the information. The uncertainties of the gridded data, and thus the uncertainty of the FAOSTAT country statistics, which are also disseminated in the dataset, depend on the spatial and temporal coverage of the GHCN stations (Menne *et al.*, 2018; Lenssen *et al.*, 2019). Furthermore, NASA-GISS recomputes the entire time series of temperature change data every year at each grid point based on the most recent set of available stations to increase the temporal comparability of the data. Measurement uncertainty is around 5–10 percent for global and regional aggregates (typically 0.1–0.2 °C), while it can be significantly higher for country values, depending on the density of the underlying network of meteorological stations. For instance, countries in regions with denser station networks, such as Northern America, Europe and Australia have smaller measurement uncertainties whereas data from countries with less dense and reliable networks, such as parts of South America, Africa and the Near East, have greater uncertainty.

A methodological note is available on the FAOSTAT temperature change on land domain (FAO. 2026).

Figure 9: 2025 network of the NASA Global historical climatology (GHCN v4)



Source: National Centers for Environmental Information. 2026. NASA Global Historical Climatology Network (GHCN v4). In: *National Oceanic and Atmospheric Administration*. [Cited March 2026]. https://www.ncei.noaa.gov/pub/data/ghcn/v4/

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